

CCS Site Installation and Configuration Guide

Release 4.2.0 (illustrative)

Notice

Notices

No CUI, classified, proprietary, or export-controlled information is contained in this document. Provided as a technical writing sample — Nathan B. Smith.

Every path, version number, command, and setting in this guide is illustrative, not the actual CCS product.

Approval

Approved for release with CCS 4.2.0 (illustrative).

Role	Name and signature	Date
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Record of changes

Revision	Date	Description	By
A	2026-09-23	Initial release for CCS 4.2.0	NBS

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Preface

Preface

Purpose, scope, audience, and conventions for this guide.

Purpose and scope

This guide tells site administrators how to install, configure, verify, and record CCS release 4.2.0 on a control-station host.

Audience

Site administrators with root access; the ISSO for witnessed steps.

Conventions

Convention	Meaning
Bold	A control on screen
Monospace	A command, file, or value you type
Line after a step	What you should see
WARNING	Risk to people or mission
CAUTION	Risk to equipment or data

Chapter 1

Installation overview

Topics:

- [References](#)

This guide takes a site administrator from verified release media to a working, tested CCS station in about 3 hours.

Every path, version number, command, and setting in this guide is illustrative, not the actual CCS product.

Item	Detail
Audience	Site administrators with root access to the control-station host
Applies to	CCS release 4.2.0, all station types
Time required	About 3 hours
People	Site administrator (performs); ISSO (witnesses and signs)

The installation at a glance



Figure 1: Five stages, each with a check that must pass

References

Documents this guide depends on, and what each one is for.

Reference	Purpose in this guide
CCS Version Description Document (VDD), release 4.2.0	Lists the exact files, versions, and checksums on the media
CCS Software User Manual, Sections 3.3 and 4.1.3	Software environment and the operator view of installation
Site configuration baseline	The approved property values for this site
DoD Instruction 5200.48, CUI	Marking of media, logs, and printed records

Chapter

2

Before you begin

Confirm every requirement before you open the media. A missing prerequisite is the most common cause of a failed install.

Requirement	Minimum (illustrative)	How to check
Operating system	RHEL 9.x, STIG-hardened site image	<code>cat /etc/redhat-release</code>
Java runtime	OpenJDK 17	<code>java -version</code>
Memory / disk	32 GB RAM; 200 GB free	<code>free -g; df -h /opt</code>
Displays	Two monitors, 1920 x 1080	Display settings
Media	Release disc, labeled and logged	Label matches the VDD



Warning: Install and verify with the datalink disconnected or in simulation mode. A station that is half-configured can send invalid commands to a live vehicle.

Chapter

3

Install and configure

Topics:

- [Verify the release media](#)
 - [Install the software](#)
 - [Configure the site](#)
 - [Verify the installation](#)
 - [Record the installation](#)
-

Verify the release media

Prove the media is exactly what the VDD says before anything touches the station.

You have the release disc and a copy of [the VDD](#). The ISSO is present.

1. Check the disc label against the VDD: release 4.2.0, build number, date, and classification marking must all match.
2. Mount the disc read-only.

```
sudo mount -o ro /dev/sr0 /mnt/ccs-media
```

3.  **CAUTION:** If any checksum fails, stop. Do not install from that media.

Verify the checksums against the list on the media.

```
cd /mnt/ccs-media
sha256sum -c SHA256SUMS
```

Every line ends in OK.

4. Compare the SHA256SUMS file's own checksum with the value printed in the VDD.
Values match exactly.
5. Record the media control number in the site media log.
The ISSO initials the entry.

The media is verified and logged.

Install the software

Run the installer from verified media and confirm the service is registered.

The media passed [Verify the release media](#) on page 18.

1. Back up the current installation, if one exists.

```
sudo /opt/ccs/bin/ccs-backup --out /var/backups/ccs-$(date +%F).tar.gz
```

2. Run the installer from the media.

```
cd /mnt/ccs-media
sudo ./install.sh --release 4.2.0 --prefix /opt/ccs
```

The installer prints `Install complete: CCS 4.2.0-1187`.

3. Confirm the service is registered but not running.

```
systemctl status ccs
```

Status shows `loaded, inactive (dead)`.

4. Unmount and remove the media; return it to storage and log the return.

Go to [Configure the site](#) on page 18.

Configure the site

Set the site properties from the approved baseline and validate the file.

Copy values from the approved site baseline; do not invent them. See [Site properties](#) on page 19 for what each property means.

1.  **CAUTION:** Leave `ccs.datalink.mode=sim` until the installation is verified. Changing it early connects the station to the live datalink.

Open `/opt/ccs/conf/site.properties` with the site-approved editor.

2. Set each property to the baseline value.

A completed file looks like this:

```
# site.properties (illustrative)
ccs.site.id=SITE-SD-02
ccs.station.type=land
ccs.platforms.enabled=mq4c,uuv
ccs.datalink.mode=sim
ccs.msg.gateway=msg-gw.site.local:7443
ccs.log.dir=/var/log/ccs
ccs.ui.banner=<per site baseline>
```

3. Validate the file.

```
sudo /opt/ccs/bin/ccs-config --check
```

Output ends with Configuration valid (7 of 7 required properties set).

Site properties

All site-specific settings live in one file, `/opt/ccs/conf/site.properties`.

Property	Example	What it actually is
<code>ccs.site.id</code>	<code>SITE-SD-02</code>	The station's unique name on the network
<code>ccs.station.type</code>	<code>land ship sub</code>	Where the station lives; sets UI layout and defaults
<code>ccs.platforms.enabled</code>	<code>mq4c,uuv</code>	Which vehicle types this station may control
<code>ccs.datalink.mode</code>	<code>sim</code>	Simulated or live vehicle connection
<code>ccs.msg.gateway</code>	<code>msg-gw.site.local:7443</code>	Address of the external messaging gateway
<code>ccs.log.dir</code>	<code>/var/log/ccs</code>	Where logs are written; must be on marked storage
<code>ccs.ui.banner</code>	Per site baseline	Classification banner text shown on every screen

Verify the installation

Start the service, run the health check, and prove the station works in simulation.

1. Start the service.

```
sudo systemctl enable --now ccs
```

2. Run the health check.

```
sudo /opt/ccs/bin/ccs-health
```

All components report UP.

3. Log in at the station with a test operator account.
The CCS window shows the classification banner and site ID.

4. Open `SMOKE-AIR.plan`, select **Validate**, then **Launch (Sim)**.
The simulated aircraft follows the route for 5 minutes with no errors.
5. Open `SMOKE-UUV.plan`, select **Validate**, then **Launch (Sim)**.
The simulated UUV completes the dive profile with no errors.
6. Required: With ISSO approval, set `ccs.datalink.mode=live` and restart the service.
Do this only after every earlier step passes.

Record the installation

An install is not complete until it is in the CM record.

1. Fill in the install record.

Field	Entry
Site ID	
Release and build installed	
Media control number	
Checksum, configuration, health, smoke test results	
Installed by (name, date)	
ISSO (name, date)	

2. File the record with the site configuration log.

Chapter

4

Upgrade and rollback

Upgrades repeat the full installation; roll back if verification fails.

Roll back if the health check or smoke test fails and cannot be fixed within the maintenance window.

1. Stop the service: `sudo systemctl stop ccs.`
2. Restore the backup made before installation.

```
sudo /opt/ccs/bin/ccs-restore --in /var/
backups/ccs-<date>.tar.gz
```

3. Start the service and repeat the health check.
4. Record the rollback and its reason in the CM log.

Appendix

A

Troubleshooting

Topics:

- [Checksum reports FAILED](#)
- [Installer reports Java 17 not found](#)
- [Configuration check names a property](#)
- [Health check shows messaging DOWN](#)
- [No classification banner on screen](#)
- [Smoke-test vehicle does not move](#)

Each problem below has its own topic: what you see, why it happens, and how to fix it.

Find your symptom in the list; if none matches, contact site CM.

Checksum reports FAILED

Checksum reports FAILED.

Damaged or wrong media.

1. Stop. Do not install.
2. Report to site CM for replacement media.

Installer reports Java 17 not found

Installer reports Java 17 not found.

Wrong or missing Java runtime.

1. Install the site-approved OpenJDK 17 package.
2. Run the installer again.

Configuration check names a property

Configuration check names a property.

Missing or misspelled value.

1. Correct the property from the site baseline.
2. Run `ccs-config --check` again.

Health check shows messaging DOWN

Health check shows messaging DOWN.

Wrong gateway address or blocked port.

1. Confirm `ccs.msg.gateway` matches the baseline.
2. Ask the network administrator to confirm port 7443 is open.

No classification banner on screen

No classification banner on screen.

`ccs.ui.banner` is empty.

1. Set `ccs.ui.banner` from the site baseline.
2. Restart the service.

Smoke-test vehicle does not move

Smoke-test vehicle does not move.

Plan not validated, or datalink not in sim mode.

1. Validate the plan.
2. Confirm `ccs.datalink.mode=sim`.

Glossary and acronyms

Terms used in this guide.

Common Control System

One common software suite that plans, controls, and manages multiple Navy unmanned vehicles.

Common Control System (CCS)

CCS

Configuration Management

Keeping a controlled, recorded baseline of exactly which versions are installed where.

Configuration Management (CM)

CM

Controlled Unclassified Information

Unclassified information that still requires safeguarding and marking under DoDI 5200.48.

Controlled Unclassified Information (CUI)

CUI

Information System Security Officer

The person responsible for a system's security posture; witnesses media checks and signs install records.

Information System Security Officer (ISSO)

ISSO

Security Technical Implementation Guide

DoD configuration standard used to harden the host operating system.

Security Technical Implementation Guide (STIG)

STIG

Version Description Document

The document that lists what is in a release: files, versions, checksums, known issues.

Version Description Document (VDD)

VDD

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